

BALANARSIMHA SANGA

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Education

National Institute of Technology Karnataka, Surathkal

Bachelor of Technology in Information Technology; CGPA: 8.67

Aug. 2023 – Present

Surathkal, Karnataka

Narayana Junior College

Telangana State Board of Intermediate Education; Percentage: 97.90%

July 2021 – May 2023

Hyderabad, Telangana

Relevant Coursework

- Data Structures and Algorithms
- Object-Oriented Programming
- Discrete Mathematics
- Operating Systems
- Database Systems
- Theory of Computation

Projects

Image Quality Manipulation using Deep Autoencoders | *Python, PyTorch*

GitHub

- Designed U-Net-based deep autoencoder architectures for image denoising and super-resolution, demonstrating consistent improvements in SSIM and PSNR on real-world datasets.
- Trained and evaluated models on both medical X-ray and natural image datasets using PyTorch and Kaggle resources, ensuring robustness across diverse image domains.
- Achieved low reconstruction errors with denoising loss of 0.00029 and super-resolution loss of 0.001, indicating high-fidelity image restoration.

Library Management System | *Node.js, Express.js, React.js, MySQL*

GitHub

- Developed a full-stack library management platform supporting end-to-end CRUD operations for users, books, borrowing transactions, and inventory records.
- Architected a relational database schema with 8 fully normalized tables, ensuring data integrity, reduced redundancy, and scalable system design.
- Built RESTful backend services and advanced SQL queries to enable efficient data retrieval, seamless frontend integration, and reliable system performance.

Online CR Election System | *Node.js, Express.js, React.js, MySQL*

GitHub

- Engineered a privacy-centric online election system supporting nomination, voting, and result phases with strict eligibility validation and controlled result disclosure.
- Implemented anonymous voting using one-time hashed tokens, transactional database locking, and voter-state verification, guaranteeing 100% prevention of duplicate votes.
- Integrated security and accountability mechanisms including OTP-based authentication, immutable audit logs, election-scoped policy acceptance, and automated notification workflows.

Metro Mate – Subway Navigation System | *C++*

GitHub

- Built a graph-based subway navigation system in C++ for Delhi, Hyderabad, and Bangalore, enabling accurate and efficient metro route planning.
- Modeled large-scale metro networks with over 280 stations using graph algorithms, demonstrating scalability for complex urban transit systems.
- Structured city-wise datasets using external data files and STL containers, enabling modular expansion and efficient data handling.
- Validated route computation logic through comprehensive testing, achieving 100% accuracy across all tested city and intercity routes.

Technical Skills

Languages: C++, C, SQL, Python, JavaScript

Developer Tools: VSCode, Google Colab, Postman, GitHub

Achievements

- Solved **300+ problems** on Codeforces — Max rating: **1300** (Profile: **Balanarsimha_S**)
- Solved **290+ problems** on LeetCode (Profile: **sangabalanarsimha**)
- Completed **Supervised Learning** module from Stanford University's Machine Learning Specialization (Prof. Andrew Ng)
- Secured All India Rank 5657 in JEE Mains 2023, ranking in the top 0.4% nationally.